



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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COURSE CODE :DSA0606

COURSE : DATA HANDLING AND VISULIZATION

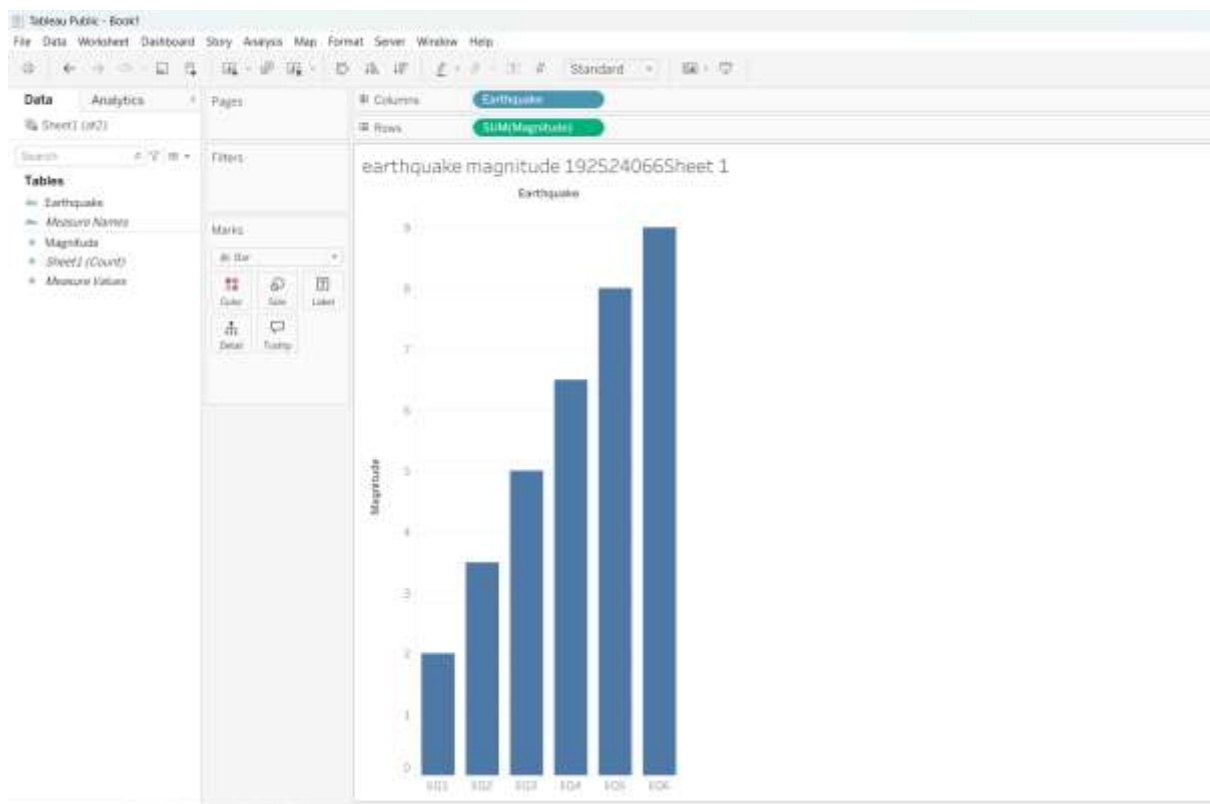
Problem 9: Earthquake Magnitude Bar Chart

Objective

To visualize earthquake magnitudes using a bar chart and understand how earthquake magnitudes are represented on a standard linear Y-axis.

Description

This visualization presents earthquake magnitudes ranging from 2.0 to 9.0 on the Richter scale. The X-axis represents different earthquakes, while the Y-axis represents their magnitude values. A bar chart is used because it allows easy comparison between the magnitudes of different earthquakes. The height of each bar corresponds to its magnitude value.



Observation

- * Earthquake magnitudes increase from 2.0 to 9.0.
- * Taller bars indicate stronger earthquakes.
- * The chart uses a linear Y-axis, so the difference between bars appears equal.
- * The visualization is simple and easy to compare, but it does not fully represent the real-world increase in earthquake energy because the Richter scale is logarithmic.

Conclusion

The bar chart effectively compares earthquake magnitudes visually. It helps identify the strongest and weakest earthquakes quickly. However, since the Richter scale is logarithmic, a small increase in magnitude actually represents a much larger increase in earthquake intensity than the chart visually suggests.

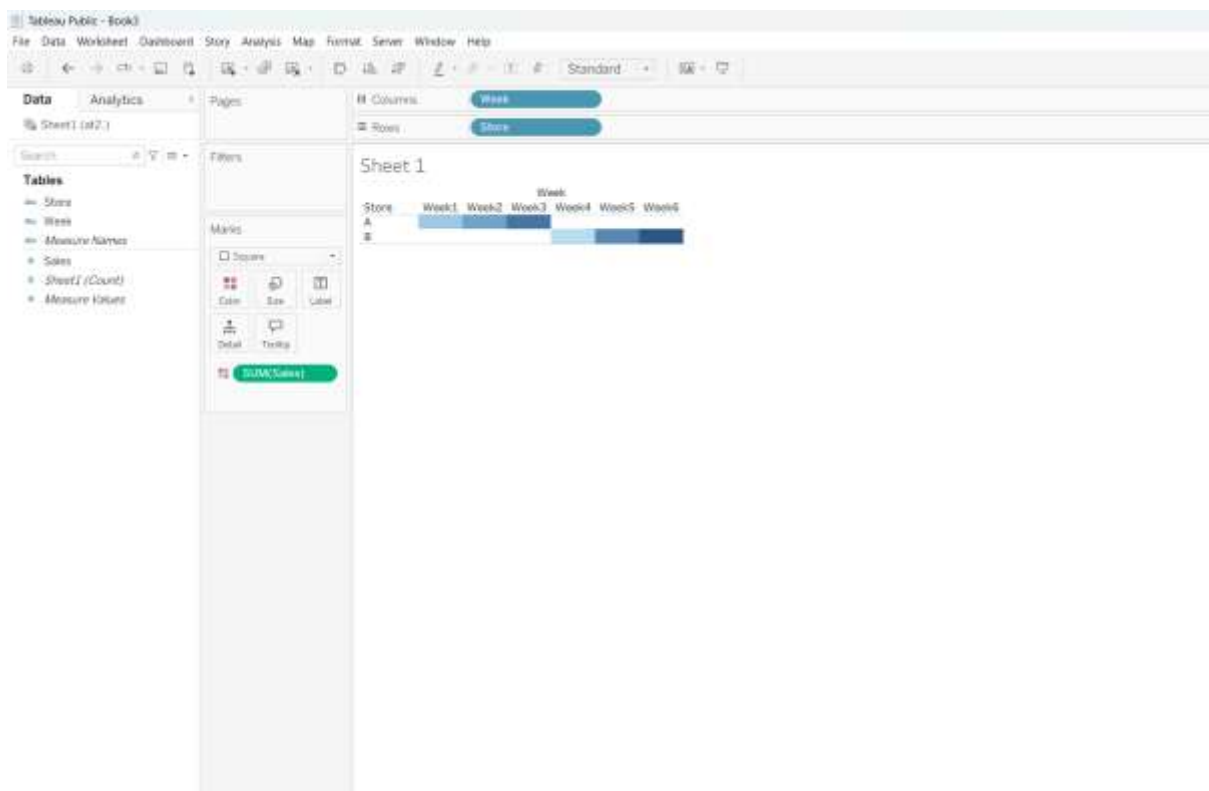
Problem 19: Weekly Sales Heatmap

Objective

To visualize weekly sales data of different stores using a heatmap with a sequential blue color scale.

Description

The heatmap displays weekly sales for multiple stores. The X-axis shows the weeks, and the Y-axis shows the stores. Sales values are represented using different shades of blue. Light blue indicates lower sales, while dark blue indicates higher sales. This visualization helps users identify sales trends without reading every numerical value.



Observation

- * Dark blue cells represent the highest sales.
- * Light blue cells represent the lowest sales.
- * Sales performance can be compared easily across different stores and weeks.
- * A darker cell immediately attracts attention and indicates better sales performance.

Conclusion

The heatmap is an effective way to compare weekly sales across stores. The sequential color scale makes it easy to identify high and low sales at a glance. This visualization improves data interpretation and helps users quickly recognize important sales patterns and trends.